Ernest Choi

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Projects

General-Purpose Processor (ALU) | *VHDL, FPGA, Quartus II*

- Designed and implemented a modular Arithmetic and Logic Unit (ALU) in **VHDL** on an **Altera FPGA**, integrating registers, a finite state machine, a 4×16 decoder, and a microcode-driven ALU core.
- Verified the design through **Quartus II** simulation, waveform analysis, and static timing verification, driving results out to 7-segment displays.
- Applied **RTL** design practice — timing analysis, modular integration, bitwise operations — to build and test digital logic at the register-transfer level.

Particle Fluid Simulation | *C++, OpenGL, GLSL, GitHub* github.com/ErChoi-com/Particle-Fluid-Simulation

- Developed a real-time fluid simulation in **C++** and **OpenGL**, writing **GLSL** shaders for particle rendering, physics-driven interaction, and GPU-accelerated parallel computation.
- Managed windowing, input, and render contexts with **GLFW** and **GLUT**, and trimmed graphics-pipeline memory allocation by 12% for smoother frames and better GPU utilization.

Experience

Technical Development Intern, City of Markham – Markham, Ontario July 2022 – August 2024

- Maintained clear **LaTeX** documentation of website workflows and operating processes, helping cut website downtime by 41%.
- Tracked *project deliverables* and reported progress to management, reducing coordination delays by 14%.
- Validated and tested system changes against quality standards and functional requirements, reducing reported errors by 20%.

Electrical Team Member, Metropolitan Aerospace Rocket Society (MARS) – Toronto, ON September 2023 – Present

- Designed and fabricated complex **PCB** circuits using **Altium** and **KiCad**, integrating custom systems for aerospace *designs* from conceptualization through *construction*.
- Enhanced power efficiency by 23% through optimization of circuit configurations, component selection, and **power distribution** across regulated voltage rails
- Performed board bring-up and debugging with **oscilloscopes**, logic analyzers, and electrical measurements to validate hardware functionality under strict performance and safety standards
- Developed embedded firmware for microcontroller-based avionics using **C/C++** and **Python**; automated testing workflows via scripting

Engineering Instructor, Obotz Robotics – Markham, Ontario September 2022 – Present

- Taught digital logic and embedded **C** programming, raising students' robotics problem-completion scores by 35% and fostering critical thinking for *Engineering* challenges.
- Coached students to excel in robotics competitions with custom **Arduino** hardware, emphasizing problem-solving and the *design* of functional systems.
- Explained students' technical progress to parents in clear, non-technical terms.

Skills

Languages: JavaScript/Node.js, Java, Python, C++, C, SQL, HTML, CSS
Frameworks/Libraries: OpenGL, WebRTC, JavaFX, React, TypeScript, Flask, FastAPI, TensorFlow
Tools: Git, PostgreSQL, Ubuntu Linux, Docker, Kafka, Tableau, Power BI, FPGA
Platforms: GitHub, GitLab, Raspberry Pi, Arduino
Relevant Courses: Digital Systems, Microprocessor Systems, Object Oriented Eng Analysis and Design, Electronic Circuits I

Education

Toronto Metropolitan University – Bachelor of Engineering in Computer Engineering 2023–2027